

SET 5: WEBSITE ANALYTICS

Dataset: Website Traffic

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

PROGRAM 1: Line Chart of Daily Page Views

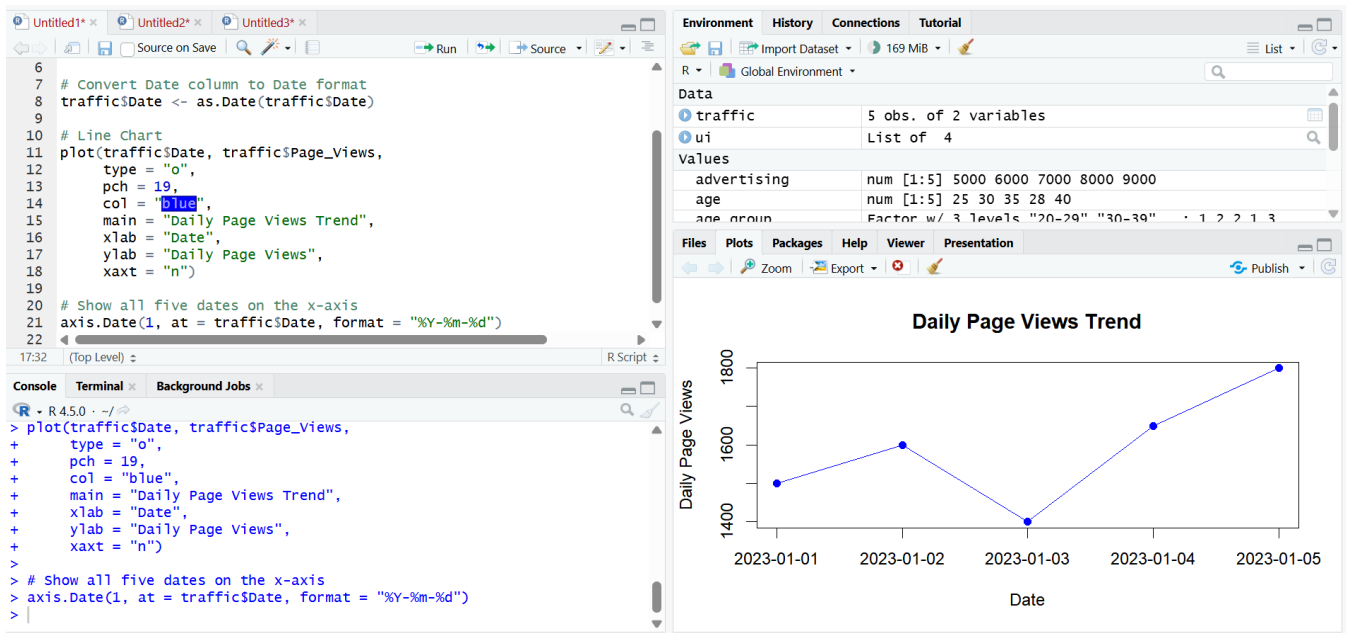
Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Dataset
traffic <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05"),
  Page_Views = c(1500, 1600, 1400, 1650, 1800)
)

# Convert Date column to Date format
traffic$Date <- as.Date(traffic$Date)

# Line Chart
plot(traffic$Date, traffic$Page_Views,
     type = "o",
     pch = 19,
     col = "blue",
     main = "Daily Page Views Trend",
     xlab = "Date",
     ylab = "Daily Page Views",
     xaxt = "n")

# Show all five dates on the x-axis
axis.Date(1, at = traffic$Date, format = "%Y-%m-%d")
```



Output 1: Daily Page Views Trend Line Chart

Observation: The line chart shows daily page views rising from 1500 on 2023-01-01 to 1600 on 2023-01-02, dropping to the lowest value of 1400 on 2023-01-03, and then increasing to 1650 on 2023-01-04 and the highest value of 1800 on 2023-01-05.

PROGRAM 2: Bar Chart of Top 3 Days by Click-through Rate

Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

Since the dataset contains only five days, N = 3 is used.

```

# Dataset
traffic <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05"),
  Click_through_Rate = c("2.3%", "2.7%", "2.0%", "2.4%", "2.6%")
)

# Convert Date column to Date format
traffic$Date <- as.Date(traffic$Date)

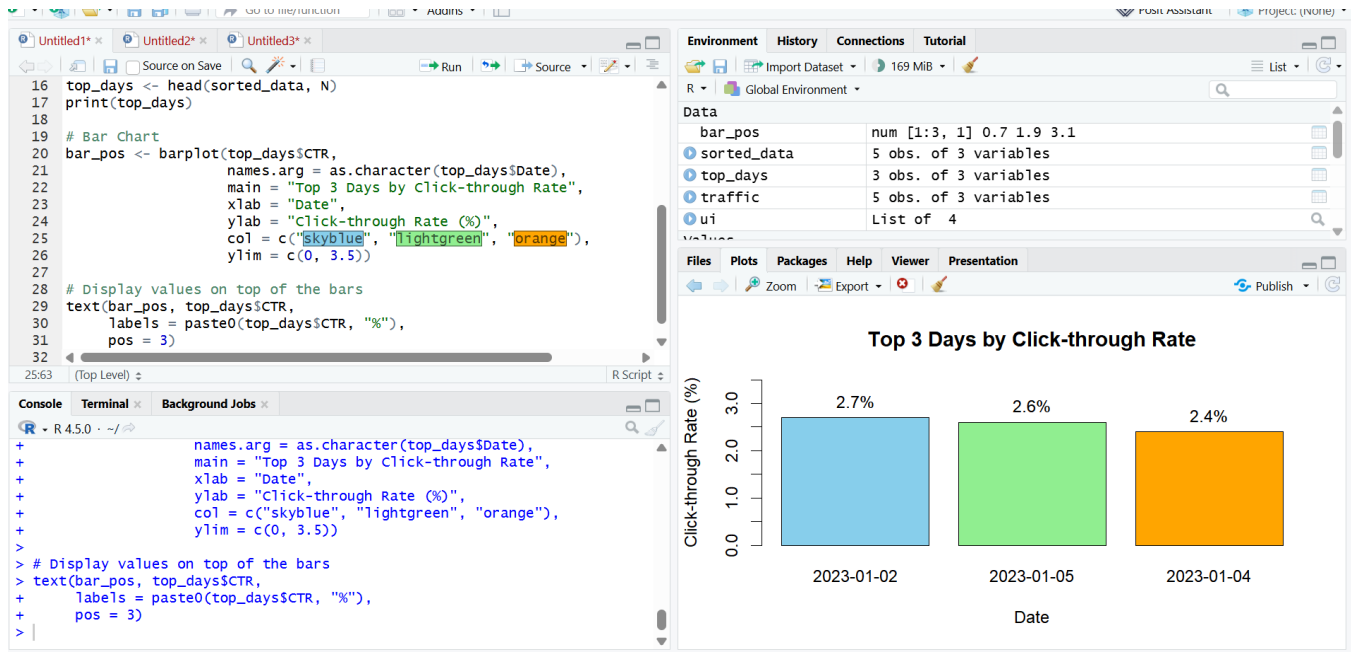
# Convert percentage text to numeric values
traffic$CTR <- as.numeric(gsub("%", "", traffic$Click_through_Rate))

# Sort in descending order and select top N days (N = 3)
N <- 3
sorted_data <- traffic[order(-traffic$CTR), ]
top_days <- head(sorted_data, N)
print(top_days)

# Bar Chart
bar_pos <- barplot(top_days$CTR,
  names.arg = as.character(top_days$Date),
  main = "Top 3 Days by Click-through Rate",
  xlab = "Date",
  ylab = "Click-through Rate (%)",
  col = c("skyblue", "lightgreen", "orange"),
  ylim = c(0, 3.5))

# Display values on top of the bars
text(bar_pos, top_days$CTR,
  labels = paste0(top_days$CTR, "%"),
  pos = 3)

```



Output 2: Top 3 Days by Click-through Rate

Observation: The three days with the highest click-through rates are 2023-01-02 (2.7%), 2023-01-05 (2.6%), and 2023-01-04 (2.4%).

PROGRAM 3: Stacked Area Chart of User Interactions

Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

Note: The original Website Traffic dataset does not contain Likes, Shares or Comments. Therefore, the following supplementary User Interaction dataset is used for this program.

Supplementary Dataset: User Interactions

Date	Likes	Shares	Comments
2023-01-01	120	50	30
2023-01-02	135	55	35
2023-01-03	110	45	28
2023-01-04	150	60	40
2023-01-05	170	70	45

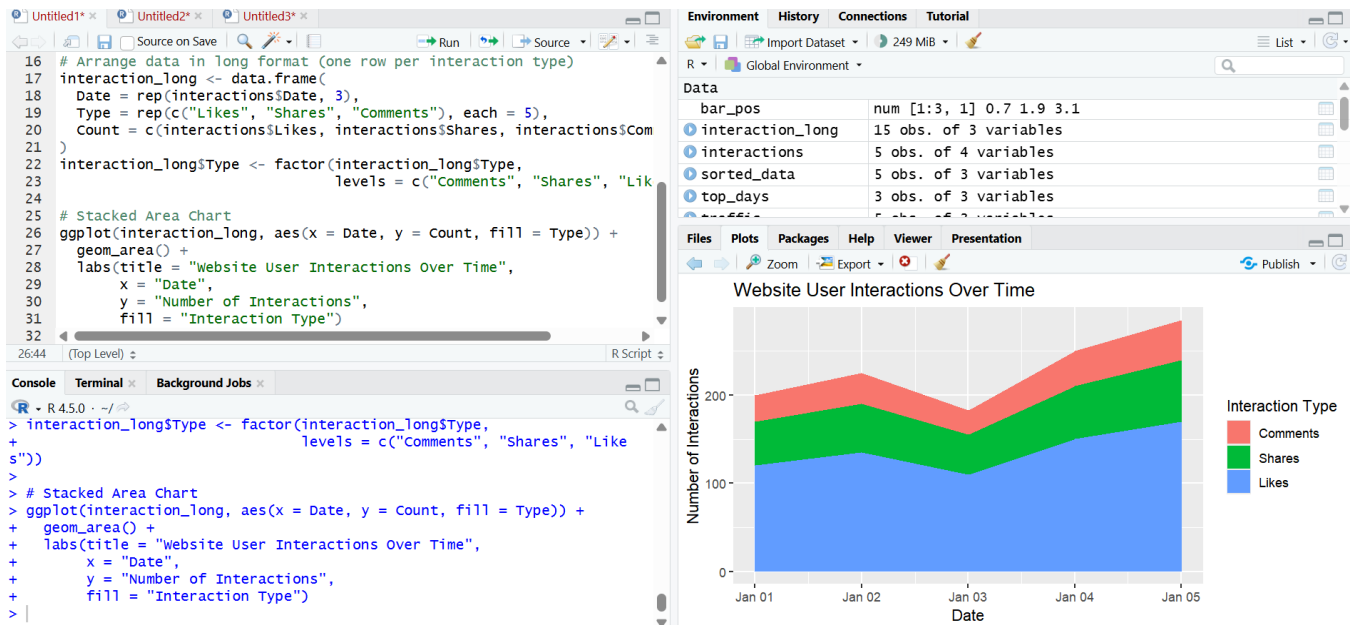
```
# Install once (if not already installed)
# install.packages("ggplot2")
library(ggplot2)

# Supplementary User Interaction dataset
interactions <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05"),
  Likes = c(120, 135, 110, 150, 170),
  Shares = c(50, 55, 45, 60, 70),
  Comments = c(30, 35, 28, 40, 45)
)

# Convert Date column to Date format
interactions$Date <- as.Date(interactions$Date)

# Arrange data in long format (one row per interaction type)
interaction_long <- data.frame(
  Date = rep(interactions$Date, 3),
  Type = rep(c("Likes", "Shares", "Comments"), each = 5),
  Count = c(interactions$Likes, interactions$Shares, interactions$Comments)
)
interaction_long$Type <- factor(interaction_long$Type,
                              levels = c("Comments", "Shares", "Likes"))

# Stacked Area Chart
ggplot(interaction_long, aes(x = Date, y = Count, fill = Type)) +
  geom_area() +
  labs(title = "Website User Interactions Over Time",
       x = "Date",
       y = "Number of Interactions",
       fill = "Interaction Type")
```



Output 3: Website User Interactions Over Time

Observation: In the supplementary dataset, Likes form the largest part of the stack on every date, followed by Shares, while Comments form the smallest part. Likes range from 110 to 170, Shares from 45 to 70, and Comments from 28 to 45. All three interaction types decrease on 2023-01-03 and reach their highest values on 2023-01-05.

PROGRAM 4: Interactive Tableau Dashboard

Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Dataset used: the original Website Traffic dataset (Date, Page Views, Click-through Rate) given above.

1. Importing the Dataset

1. Save the Website Traffic dataset as an Excel or CSV file (for example, website_traffic.xlsx).
2. Open Tableau and, under Connect, select Microsoft Excel (or Text file for CSV).
3. Browse to the saved file and open it. Drag the sheet to the canvas.
4. In the Data Source tab, check that Date is recognized as a Date field, Page Views as a numeric field, and Click-through Rate as a numeric/percentage field. Change the data type by clicking the icon above the column name if required.
5. Click Sheet 1 to start creating the charts.

2. Create the Line Chart

1. Rename the worksheet to "Daily Page Views".
2. Drag Date to Columns (right-click it and choose Exact Date).
3. Drag Page Views to Rows.
4. In the Marks card, select Line as the mark type. Add Circle marks or show mark labels (Label > Show mark labels) if required.
5. Double-click the sheet title and enter "Daily Page Views Trend". Make sure the axis titles read Date and Page Views.

3. Create the Bar Chart

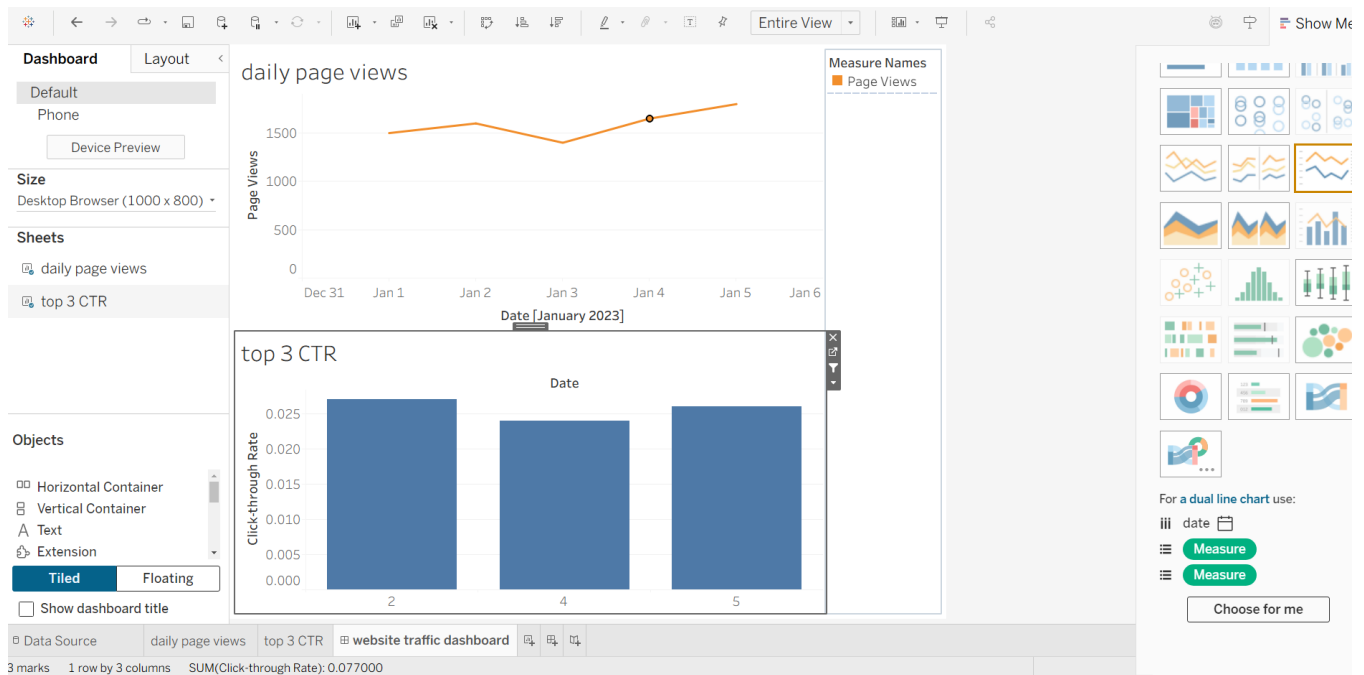
1. Add a new worksheet and rename it to "Top 3 Click-through Rates".
2. Drag Date to Rows and Click-through Rate to Columns.
3. Click the Sort Descending button on the toolbar to sort the bars by Click-through Rate.
4. Drag Date to Filters, open the Top tab, select By field, choose Top 3 by Click-through Rate (Sum), and click OK.
5. In the Marks card, select Label > Show mark labels to display the percentage values on the bars.
6. Set the sheet title to "Top 3 Days by Click-through Rate". Make sure the dates and percentage values are clearly visible.

4. Create the Dashboard

1. Click New Dashboard and rename it to "Website Traffic Dashboard".
2. Drag the "Daily Page Views" worksheet into the dashboard area.
3. Drag the "Top 3 Click-through Rates" worksheet into the dashboard area below or beside the line chart.
4. Resize and arrange both charts neatly, and add a dashboard title such as "Website Traffic Dashboard" (Dashboard > Show Title).

5. Add Interactivity

1. Select the "Top 3 Click-through Rates" chart on the dashboard and click the Use as Filter icon (funnel symbol) at the top right of the chart.
2. Add a Date filter: on the "Daily Page Views" worksheet, drag Date to Filters, choose a range of dates, then use Show Filter so that the date control appears on the dashboard.
3. Click a bar (date) in the bar chart: the line chart updates to show the page views for the selected date. Click a blank area to clear the selection.
4. Use the Date filter on the dashboard to restrict the line chart to a chosen range of dates and explore the corresponding traffic.



Output 4: Tableau Website Traffic Dashboard

Observation: The dashboard shows the daily page views trend and the top 3 click-through rate days side by side. The user can select a bar or apply the Date filter to explore the website traffic information for the chosen dates.

Procedure

1. Create the Website Traffic data frame in R containing Date, Page Views and Click-through Rate.
2. Convert the Date column into Date format using `as.Date()`.
3. Create the line chart using Date and Page Views with `plot()`.
4. Sort the Click-through Rate values using `order()` and create the Top 3 bar chart using `barplot()`.
5. Create the supplementary User Interaction dataset containing Likes, Shares and Comments.
6. Create the stacked area chart using `ggplot2` and `geom_area()`.
7. Save the Website Traffic dataset as an Excel/CSV file and import it into Tableau.
8. Create the Daily Page Views line chart in Tableau.
9. Create the Top 3 Click-through Rate bar chart in Tableau using a Top N filter (N = 3).
10. Combine both worksheets into the Website Traffic Dashboard.
11. Add filters/interactivity to the dashboard using Use as Filter and a Date filter.
12. Test the dashboard by selecting different dates/bars and observing the chart updates.

Result

The Website Traffic data was successfully visualized using a line chart, a Top 3 bar chart, a stacked area chart, and a Tableau interactive dashboard.

The line chart was used to visualize daily page views over time. The bar chart was used to identify the top 3 days with the highest click-through rates. The stacked area chart was used to display the distribution of Likes, Shares and Comments using the supplementary interaction dataset. The Tableau dashboard was created for interactive exploration of website traffic data.